

Energy-efficient Analog Integrated Circuits

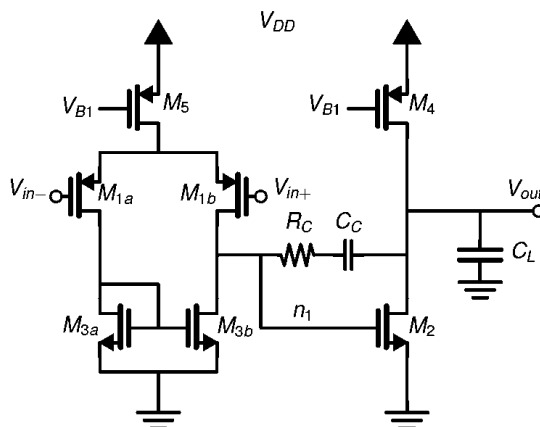
Sensor Interface – Exercise 1

Assignment

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1. Consider the design strategy described in Section 6 of the tutorial. Repeat the analysis for an n-channel device `nch_hvt`. Assume $L = 330$ nm and a drain-source voltage $V_{DS} = 0.6$ V. Design the input pair to achieve $g_m = 10$ mS in the case of $g_m/I_D = 25$, $g_m/I_D = 10$, and $g_m/I_D = 7.5$. Hint: you can now sweep the gate voltage V_G and fix the source voltage V_S .
2. The schematic of a two-stage operational transconductance amplifier is shown below. Assume the amplifier operates with a supply $V_{DD} = 1.2$ V and the input DC level is $V_{DD}/2$. Design the amplifier to meet the following specifications and explain your design strategy carefully:
 - Unity-gain frequency $f_u > 1$ MHz
 - Open-loop gain $A_{OL} > 45$ dB
 - Load capacitance $C_L = 5$ pF
 - Phase Margin $PM > 64^\circ$
 - Power consumption < 20 μ W
 - Bias M_{1a} , M_{1b} and M_2 to operate in weak inversion: $g_{m1}/I_{D1} = g_{m2}/I_{D2} = 20$
 - Minimize the systematic input offset
 - Assume $R_C = 1/g_{m2}$



3. What is the effect of the compensation resistance R_C on the circuit transfer function? Can you improve the phase margin by using a different value of R_C ?
4. Run a noise analysis and simulate the input-referred noise. Modify the design targeting an input-referred noise lower than $70 \text{ nV}/\sqrt{\text{Hz}}$ at 1 kHz (you don't need to satisfy the specifications from 2).
5. Simulate the noise efficiency factor of the two-stage amplifier.